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Project Number: #4

Project Name: Vectorized Array Multiplication and Multiplication/Reduction using SSE

Project Report #4

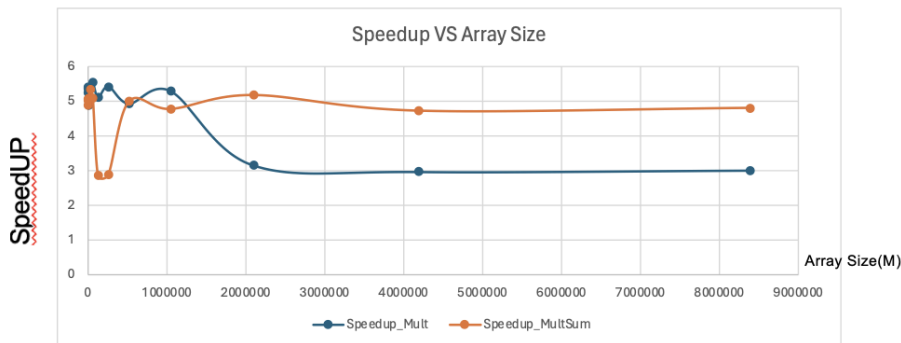
1. What machine you ran this on

MacBook Air (M2 chip)

2. Show the 2 tables of performances for each array size and the corresponding speedups

ArraySize	NonSIMD_Mult	SIMD_Mult	Speedup_Mult	NonSIMD_MultSum	SIMD_MultSum	Speedup_MultSum
1024	123.97	650.98	5.25	131.91	645.66	4.89
2048	124.6	667.98	5.36	143.56	720.62	5.02
4096	132.8	720.11	5.42	143.87	729.87	5.07
8192	144.64	781.31	5.4	166	843.23	5.08
16384	164.38	801.68	4.88	188.33	961.28	5.1
32768	195.56	1050.69	5.37	221.56	1185.1	5.35
65536	236.27	1309.8	5.54	354.49	1805.7	5.09
131072	328.46	1677.87	5.11	354.57	1014.72	2.86
262144	308.54	1673.06	5.42	343.3	990.98	2.89
524288	323.51	1595.55	4.93	352.05	1758.91	5
1048576	286.84	1518.39	5.29	351.74	1682.19	4.78
2097152	298.16	942.21	3.16	329.42	1705.55	5.18
4194304	304.55	904.51	2.97	339.44	1604.25	4.73
8388608	301.68	908.32	3.01	338.04	1627.58	4.81

3. Show the graphs (or graph) of SIMD/non-SIMD speedup versus array size (either one graph with two curves, or two graphs each with one curve)



4. What patterns are you seeing in the speedups?

Speedup_Mult (Blue Line): This line represents the speedup achieved by using SIMD for multiplication ($C[i] = A[i] * B[i]$). The speedup increases sharply as the array size increases, peaking around an array size of 300,000 with a speedup factor of approximately 5. After this peak, the speedup decreases slightly and then stabilizes around a speedup factor of about 3.

Speedup_MultSum (Orange Line): This line represents the speedup achieved by using SIMD for $\sum A[i] * B[i]$. The speedup increases sharply with the array size, peaking just below a speedup factor of 5 at an array size of about 300,000. After this point, the speedup remains relatively constant at 4.9, indicating that the SIMD operations maintain a consistent performance advantage for larger array sizes.

5. Are they consistent across a variety of array sizes?

According to my graph above that they are not entirely consistent across a variety of array sizes.

6. Why or why not, do you think?

Because for smaller array sizes, the speedup increases sharply. This is because SIMD operations process multiple data points simultaneously, leading to a significant performance improvement over non-SIMD operations that process data sequentially. But after 200,000 the speed up stableizes.